Requirement Analysis of PeerGrid: An Integrated Academic Networking and College Guidance Platform

TEAM 2

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***Abstract*—Requirement analysis is a fundamental phase of the Software Development Life Cycle (SDLC) in which the functional and technical expectations of a software system are identified, analyzed, and documented before implementation. This paper presents the requirement analysis for PeerGrid, an integrated digital platform that combines social networking, academic collaboration, college comparison, KEAM admission guidance, and campus event management within a single application. The analysis identifies the functional, non-functional, technical, and system requirements necessary to ensure secure student verification, reliable institutional information, and intelligent recommendation services. The findings provide a foundation for the design, implementation, and future scalability of the proposed system.**

***Keywords*—requirement analysis; functional requirements; non-functional requirements; system architecture; academic networking platform; KEAM college predictor**

I. Introduction

Requirement analysis is a fundamental phase of the Software Development Life Cycle (SDLC) in which the functional and technical expectations of a software system are identified, analysed, and documented before implementation. A well-defined requirement analysis reduces development risks, minimizes ambiguity, and provides a clear roadmap for system design and implementation. For the proposed PeerGrid platform, the analysis focuses on identifying the needs of prospective students, enrolled students, college administrators, and system administrators.

PeerGrid is designed as an integrated digital platform that combines social networking, academic collaboration, college comparison, KEAM admission guidance, and campus event management within a single application. Unlike conventional social networking platforms, the proposed system emphasizes secure student verification, reliable institutional information, and intelligent recommendation services. Therefore, the system must satisfy not only functional requirements but also stringent technical, security, performance, and scalability requirements to ensure long-term reliability.

II. Functional Requirements

The functional requirements define the services and operations that PeerGrid must perform. The system shall provide a secure user registration and authentication mechanism using institutional email verification, encrypted password storage through BCrypt, JSON Web Token (JWT) based session management, and role-based access control for students and administrators. Each authenticated user shall be able to create and manage a personalized academic profile containing educational information, technical skills, achievements, interests, and privacy preferences.

The platform shall maintain a centralized college repository containing verified information regarding engineering colleges in Kerala, including accreditation, academic programs, placement statistics, infrastructure, research activities, hostel facilities, and campus achievements. An intelligent college ranking module shall evaluate institutions using multiple weighted parameters such as academic excellence, placement performance, research output, student satisfaction, infrastructure quality, and extracurricular activities.

The KEAM College Predictor shall analyse historical admission cutoff datasets together with the student's entrance rank, reservation category, preferred engineering branch, and district preference to generate admission probability predictions. PeerGrid shall also provide a secure academic social networking environment where verified students can communicate through private messaging, discussion forums, multimedia posts, comments, and collaborative communities. In addition, the system shall support event publication, online registration, automated reminders, attendance management, verified reviews, notification services, and a centralized administrative dashboard for user management, content moderation, analytics, and report generation.

III. Non-Functional Requirements

Non-functional requirements specify the quality characteristics of the proposed system. PeerGrid shall deliver high performance by maintaining API response times below two seconds under normal operating conditions and supporting thousands of simultaneous users through optimized database indexing, caching mechanisms, asynchronous processing, and efficient RESTful API design.

Security is considered one of the primary quality attributes of the system. All communications between client and server shall be protected using HTTPS with TLS encryption. User passwords shall never be stored in plain text and must be hashed using BCrypt. The application shall implement JWT authentication, role-based authorization, input validation, Cross-Site Scripting (XSS) protection, Cross-Site Request Forgery (CSRF) prevention, NoSQL injection prevention, secure file upload validation, and comprehensive audit logging.

The system shall maintain high reliability with automated database backups, fault recovery mechanisms, centralized exception logging, and cloud-based deployment capable of providing high availability. Maintainability shall be achieved through modular architecture, reusable components, proper documentation, version control using Git, and automated testing. Furthermore, the user interface shall remain responsive across Android, iOS, tablets, and modern web browsers while complying with usability and accessibility principles.

IV. Technical and System Requirements

The proposed system follows a three-tier architecture consisting of the presentation layer developed using Flutter, the business logic layer implemented with Node.js and Express.js, and the database layer managed through MongoDB Atlas. Communication between the client and server shall be performed through RESTful APIs using JSON data interchange, while Firebase Cloud Messaging shall be used for real-time notifications.

The development environment requires a workstation equipped with at least an Intel Core i5 or AMD Ryzen 5 processor, 8 GB RAM, and 256 GB SSD storage. The production server should operate on Ubuntu Linux with a quad-core processor, 16 GB RAM, SSD storage, Docker container support, Nginx as a reverse proxy, and cloud deployment infrastructure.

The software stack includes Flutter SDK, Dart, Node.js, Express.js, MongoDB Atlas, Firebase Authentication, Firebase Cloud Messaging, Git, Android Studio or Visual Studio Code, Postman, Docker, and Nginx. The database shall implement schema validation, indexing, backup policies, soft deletion, and optimized query execution. The backend shall provide versioned REST APIs with standardized HTTP response codes, JWT authentication, API rate limiting, and comprehensive API documentation using the OpenAPI/Swagger specification. Continuous monitoring, centralized logging, crash analytics, and performance metrics shall be incorporated to ensure operational stability and simplify future maintenance.

V. Conclusion

The requirement analysis demonstrates that PeerGrid is designed as a comprehensive, secure, and scalable college connectivity platform rather than a conventional social networking application. By integrating intelligent college recommendation, verified student networking, event management, academic collaboration, and cloud-based services into a unified architecture, the system addresses the limitations of existing educational platforms. The functional, non-functional, technical, and system requirements presented in this paper provide a strong foundation for system design, implementation, testing, deployment, and future scalability while ensuring security, maintainability, and high performance.

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VI. Literature Survey

Educational technology has transformed the way students gather information about colleges and communicate with peers. Although numerous online platforms provide admission guidance, college rankings, discussion forums, and professional networking, most of them focus on only one aspect of the student journey. Very few platforms combine verified student communities, college comparison, admission prediction, event discovery, and campus networking into a unified ecosystem. The purpose of this literature survey is to study existing research and commercial solutions, identify their strengths and limitations, and establish the research gap that motivates the development of PeerGrid.

A. Review of Research Papers

1) Social Networking in Higher Education: This work explains how social networking platforms improve communication, collaborative learning, and student engagement in universities. However, it primarily focuses on communication and learning and does not provide admission guidance, college ranking, or verified student-only communities.

2) Recommendation Systems for Higher Education: This study examines recommendation algorithms that help students choose institutions based on academic performance and preferences. The recommendation approach is based mainly on academic data and lacks community interaction and real-time student feedback.

3) Educational Data Mining for Student Decision Support: This paper demonstrates how historical educational datasets can be used to assist students in academic planning. The scope is limited to analytics and prediction without social networking features.

4) Online Communities and Collaborative Learning: This work highlights the importance of peer discussions, knowledge sharing, and collaborative learning among university students, but does not address institutional comparison or college selection.

5) Machine Learning Based College Recommendation: This study uses historical admission data to predict suitable colleges for students. It focuses only on prediction and ignores campus life, events, reviews, and student networking.

6) Mobile Learning Communities: This paper explores mobile-first educational communities that increase accessibility and engagement, but provides learning support without admission and college comparison modules.

7) Trustworthy Review Systems: This work discusses mechanisms for collecting authentic user reviews while reducing fake content, though it is not tailored to higher education ecosystems.

8) Campus Event Management Platforms: This study examines digital solutions for managing student events and registrations, limited to event management without social networking or recommendation features.

9) Student Information Portals: This paper reviews systems that centralize academic information and announcements. Such systems are usually institution-specific and not designed for statewide comparison.

10) AI-driven Personalized Education Platforms: This work describes personalization techniques using AI and user behavior, focusing on learning content rather than college admissions and peer networking.

B. Existing Solutions

Table I summarizes commercial and community-driven platforms that currently serve overlapping aspects of the student journey, along with their respective strengths and limitations.

TABLE I. Comparison of Existing Platforms

Platform	Strengths	Limitations
Shiksha	College information, rankings, admission guidance	Limited peer interaction and no verified student community.
Collegedunia	Reviews, rankings, admission information	No campus social networking or KEAM-focused recommendation.
LinkedIn	Professional networking	Not designed for campus communities or admissions.
Facebook Groups	Large communities and event sharing	Verification issues and scattered information.
Unstop	Competitions, internships, events	No college comparison or admission guidance.
Reddit	Peer discussions and experiences	Unverified opinions and lack of structured college data.

C. Technologies Used in Existing Systems

Modern educational platforms commonly employ React, Angular, Flutter, Node.js, Django, Spring Boot, Firebase Authentication, MySQL, PostgreSQL, MongoDB, REST APIs, cloud computing services, push notification frameworks, and machine learning algorithms for recommendation and personalization.

D. Limitations of Existing Systems

Despite significant technological advancements, existing platforms exhibit several shortcomings. They generally separate admission guidance, social networking, college reviews, event management, and communication into different applications. Most platforms do not verify whether users actually belong to the colleges they discuss, resulting in unreliable reviews and misinformation. Furthermore, there is no comprehensive platform dedicated to engineering colleges in Kerala that integrates KEAM counselling support with authentic student interaction.

E. Research Gap

The literature survey indicates the absence of a unified platform that combines verified student networking, dynamic college rankings, historical KEAM-based admission prediction, authentic reviews, event management, and college-specific communities. Existing systems either provide information without interaction or interaction without reliable academic decision support.

F. How PeerGrid Addresses the Research Gap

PeerGrid addresses these shortcomings by integrating multiple services into one application. The proposed system provides verified student registration, Kerala-focused college directories, multi-parameter college rankings, KEAM college prediction using historical cut-off data, campus discussion forums, event management, authentic reviews, secure messaging, and personalized recommendations. By combining academic decision support with an exclusive student social network, PeerGrid offers a comprehensive ecosystem for prospective and current college students.

G. Chapter Summary

The literature survey demonstrates that although many educational platforms and research studies contribute valuable features, none provide an integrated Kerala-centric solution that combines admission assistance, verified networking, and comprehensive college comparison. These findings justify the development of PeerGrid as a novel platform capable of supporting students throughout their admission journey and campus life.

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H. Comparative Analysis of Existing Systems

A comparative analysis of existing educational and social networking platforms reveals that most systems address only a limited portion of the student lifecycle. Some platforms provide admission guidance, while others emphasize networking or professional development. None of the reviewed systems provide a unified platform specifically designed for Kerala engineering students admitted through KEAM. Table II summarizes this feature-by-feature comparison.

TABLE II. Feature-Wise Comparison of Existing Platforms and PeerGrid

Feature	Shiksha	Collegedunia	LinkedIn	Facebook	Unstop	Reddit	PeerGrid
College Information	✓	✓	✗	✗	✗	Partial	✓
KEAM Prediction	✗	✗	✗	✗	✗	✗	✓
Verified Student Community	✗	✗	✗	✗	✗	✗	✓
College Ranking	✓	✓	✗	✗	✗	✗	✓
Campus Events	Partial	Partial	✗	✓	✓	Partial	✓
Student Reviews	✓	✓	✗	Unverified	✗	Unverified	Verified
Private Messaging	✗	✗	✓	✓	✗	Partial	✓
Discussion Forums	Partial	Partial	✗	✓	✓	✓	✓

I. Research Gap Analysis

The literature survey clearly indicates that no existing platform integrates admission support, verified campus networking, college comparison, event discovery, and authentic student feedback into a single ecosystem.

Most admission portals primarily focus on publishing institutional information, rankings, and eligibility criteria. They rarely provide direct interaction between prospective students and current students.

General-purpose social networking platforms allow communication but lack mechanisms for verifying college affiliation, resulting in misinformation and unreliable reviews.

Professional networking platforms are designed for career development rather than student life and do not provide admission assistance or campus-specific services.

Another significant gap is the absence of a Kerala-centric solution that considers KEAM admission trends, historical cut-off ranks, and branch-wise seat availability while simultaneously supporting student collaboration.

J. Proposed Solution — PeerGrid

PeerGrid is proposed as a comprehensive college connectivity platform exclusively for higher education institutions in Kerala. The platform combines academic decision support with secure student networking. It enables prospective students to identify suitable colleges based on KEAM rank while allowing current students to communicate through verified college communities. The proposed solution provides the following capabilities:

- Verified student registration using institutional details.
- Comprehensive directory of Kerala engineering colleges.
- Dynamic ranking based on academics, placements, infrastructure, events, clubs, research, sports and student feedback.
- KEAM College Predictor using historical admission cut-off trends.
- College-specific social feeds, messaging and discussion forums.
- Event management with registrations and reminders.
- Authentic student reviews for academics, hostel, faculty and campus life.
- Personalized recommendations using user interests and academic preferences.
- Administrative dashboard for moderation, analytics and verification.

K. Advantages of the Proposed System

PeerGrid offers several advantages over existing systems. It reduces the dependence on multiple websites by providing admission guidance, networking, reviews and event management in one application. The use of verified student accounts improves the authenticity of reviews and discussions. Its Kerala-focused design makes it highly relevant for KEAM aspirants while its scalable architecture allows future expansion to other universities and states.

L. Chapter Summary

The literature survey and comparative analysis demonstrate the necessity of a unified educational platform that supports the complete student journey. PeerGrid addresses the identified research gaps by integrating verified student networking, intelligent admission guidance, dynamic college ranking, event management, and collaborative discussion into a single secure platform.

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